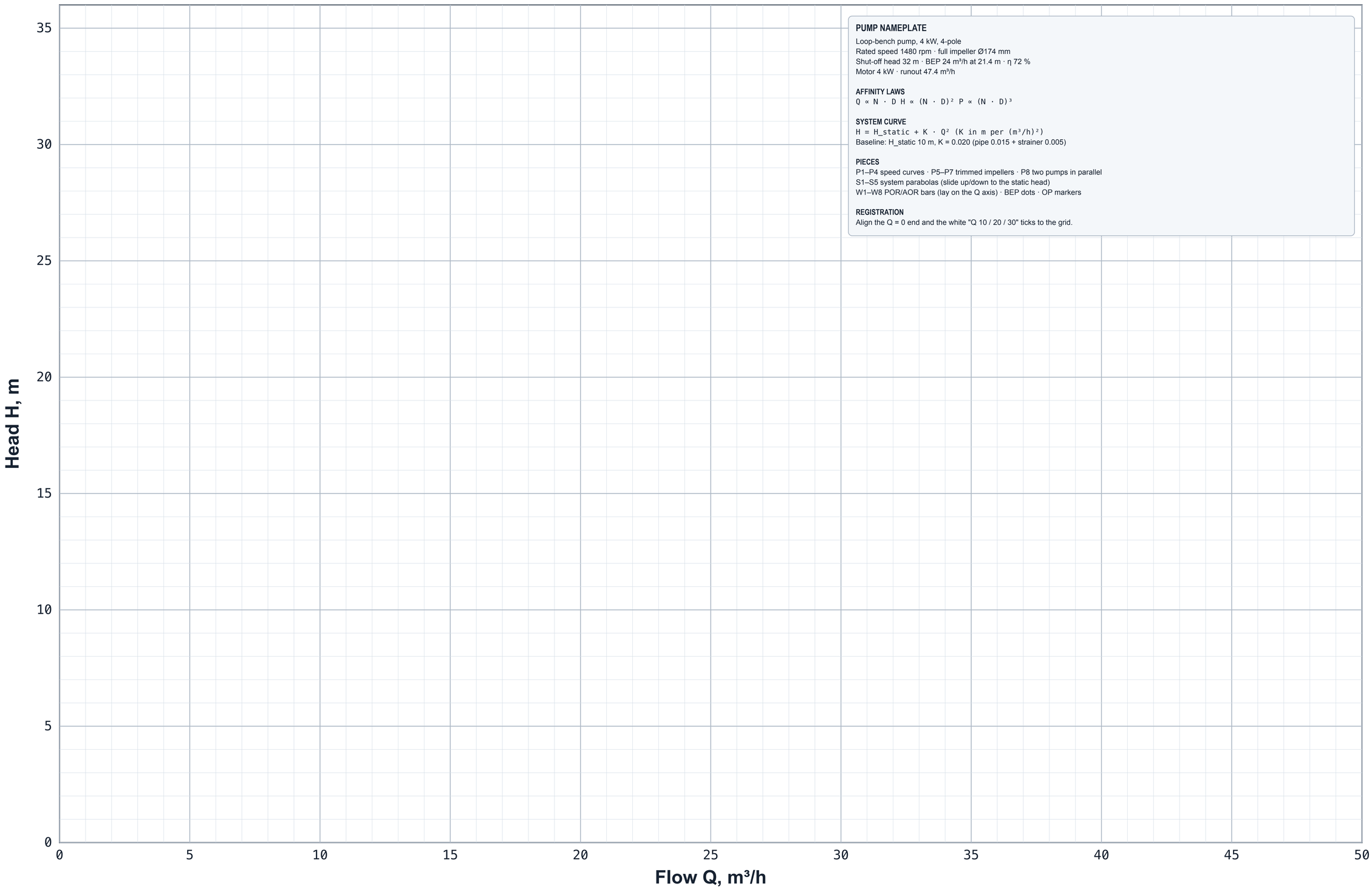
Magnetic Operating-Point Board

Centrifugal pump H–Q chart · place the pump curve, the system curve and the POR/AOR bar, then mark the operating point

Mechanical engineering training · Istec Training

[Illustrative pump and system data, for training only]



PUMP NAMEPLATE

Loop-bench pump, 4 kW, 4-pole
Rated speed 1480 rpm · full impeller Ø174 mm
Shut-off head 32 m · BEP 24 m³/h at 21.4 m · η 72 %
Motor 4 kW · runout 47.4 m³/h

AFFINITY LAWS

$Q \propto N \cdot D \quad H \propto (N \cdot D)^2 \quad P \propto (N \cdot D)^3$

SYSTEM CURVE

$H = H_{static} + K \cdot Q^2$ (K in m per $(m^3/h)^2$)
Baseline: H_static 10 m, K = 0.020 (pipe 0.015 + strainer 0.005)

PIECES

P1–P4 speed curves · P5–P7 trimmed impellers · P8 two pumps in parallel
S1–S5 system parabolas (slide up/down to the static head)
W1–W8 POR/AOR bars (lay on the Q axis) · BEP dots · OP markers

REGISTRATION

Align the Q = 0 end and the white "Q 10 / 20 / 30" ticks to the grid.

Static head: the bottom-left end of a system piece sits on the H = static head line at Q = 0. POR = 70–120 % of Q_BEP (ANSI/HI 9.6.3) · AOR = 50–130 % (illustrative) · both scale with speed and diameter.

Print at 100 % on A1 (841 × 594 mm). Mount on a steel sheet or a magnetic whiteboard. Pieces are on the matching A1/A3 sheets: laminate, cut on the black outline, magnetic tape on the back.